

# Myrrh Khan

[myrrhkhan.github.io](https://myrrhkhan.github.io) | [linkedin.com/in/myrrh-khan/](https://linkedin.com/in/myrrh-khan/)

## SUMMARY

CS and Robotics graduate with hands-on experience delivering solutions to real-world stakeholders. Experienced with embedded systems, computer vision, and full-stack development.

## SKILLS

*Languages:* C/C++, Rust, Java, Python, SQL (Postgres, Oracle, SQLite), JavaScript/TypeScript (SvelteJS, React)

*Tools, Libraries, Other:* AWS, Google Cloud Platform, PyTorch, Flask, Agile (SCRUM)

*Awards:* **1st Place** at Draper Capstone Symposium, **Finalist** at WPI RBE Project Presentation Day, GoatHacks 2024 **Honorable Mention**

## EDUCATION

### Worcester Polytechnic Institute

Worcester, MA

*BS in Computer Science, Minor in Robotics Engineering | GPA: 3.63/4.0*

*August 2022 – May 2025*

- **Relevant Courses:** Deep Learning for Perception, Software Engineering, Mobile and Ubiquitous Computing, Distributed Systems, Advanced Computer Networks, Unified Robotics II
- **Extracurriculars:** WPI Orchestra (bassist) and Pep Band, Alpha Phi Omega Coed Service Fraternity (Historian, Weekly Service Chair)

## EXPERIENCE

### Agentic Workflow Engineer Intern

Alexandria, VA

*MARi*

*May 2025 – August 2025*

- Created and presented MVP for AI-powered career acceleration platform for companies with **1000+ employees**
- Integrated React frontend with Flask server and PostgreSQL database deployed in Google Cloud via **Kubernetes** and Docker

### UAV-UGV Collaboration Researcher

Cambridge, MA

*Draper*

*August 2024 – May 2025*

- Developed C++ firmware for ESP-32 microcontrollers with integrated gyroscope sensors via PlatformIO
- Designed system with 4-person team to reliably and autonomously land and charge a drone onto a ground vehicle via AprilTags. **93% success rate**
- Achieved **50x** UAV battery life increase, expanding feasibility for UAV use in **search and rescue operations**
- **Won 1st place** at Draper Capstone Symposium, and **reached finals round** at WPI Project Presentation Showcase

### Software Developer, Researcher

Worcester, MA

*WPI RoboCareLab*

*October 2024 – May 2025*

- Built Flask web portal enabling Pepper robots to access patient memory data for **improved dementia care**

## PROJECTS

### Violet: An Award-Winning Handwriting to Text Converter App

*Rust, React, SvelteJS, Python, Tensorflow, AWS, GitHub Actions*

*January 2024 – Present*

- <https://devpost.com/software/violet-yrafct>
- Built full-stack handwriting-to-text app in **48 hours**, achieving **67% accuracy**, enhancing class note-taking and winning **GoatHacks 2024 Honorable Mention**, now deploying via **AWS Lambda**

### RBE 474x Projects

*PyTorch, Blender*

*September 2024*

- Created **diffusion model from scratch** to AI-generate images for dataset, deployed training onto high-performance computing clusters via **SLURM**
- Generated images to **hijack depth-perception models**
- Generated augmentations for dataset built for drone-window navigation in Blender, achieved **80% accuracy**

### Android Face Tracker

*Kotlin, Python, MediaPipe, Flask, Jetpack*

*May 2025*

- **Innovated** upon Apple's Center Stage **face-tracking** feature by porting it to Android
- Constructed intuitive Material 3 UI with Kotlin and Jetpack